JTMK Project Report:

<u>Smart Parking Detection System</u>

Project Title:
Smart Parking Detection System with Real-Time Display

Team Members:
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Introduction:
The Smart Parking Detection System is designed to detect parking space availability using ultrasonic sensors and to display it on an LCD screen. This system models an efficient solution for managing limited parking spaces in real-time.

Objectives:

1. To detect vehicle presence using ultrasonic sensors.
2. To display space availability on an LCD.
3. To reduce search time for drivers in parking lots.
4. To practice sensor logic and display integration.

Tools and Technologies Used:

• Hardware: Arduino Uno, Ultrasonic Sensors (HC-SR04), LCD Display, LEDs

• Software: Arduino IDE, Fritzing

• Programming Language: C/C++

Features:

• Real-time vacancy status updates

• LCD display with slot numbers

• LED indicators for available/occupied slots

• Simple and cost-effective design

Circuit Design Diagram:

(Insert image here of the full system circuit)

Implementation Process:

1. Planning: Defined parking logic and LCD layout.
2. Hardware Assembly: Placed sensors in each slot and connected display.
3. Programming: Developed loop for sensor reading and display refresh.
4. Testing: Simulated different vehicle positions.
5. Improvements: Fine-tuned sensor distance detection.

Results:

• Detected cars within 10–15 cm range

• Instant LCD update (<2 sec delay)

• Accuracy rate: 95% indoors

Challenges:

• Ultrasonic signal interference between sensors

• LCD flickering due to rapid updates

• Compact wiring in small space

Future Enhancements:

• Add mobile app for remote view

• Integrate with barrier control system

• Use IoT for cloud-based tracking

Conclusion:

This project demonstrated how sensor-based monitoring can improve real-life scenarios like parking. It served as a strong foundation for smart city concepts.

References:

1. HC-SR04 Datasheet – https://components101.comDHT11 Sensor Guide – https://randomnerdtutorials.com

2. Arduino Parking System Projects – https://electronicshub.org
3. LCD with Arduino Tutorial – https://tutorialspoint.com